

EDUCATION

- George Washington University.** *Ph.D. in Systems Engineering.* May 2023
Advisor: Prof. [Thomas Mazzuchi](#) and Prof. [Shahram Sarkani](#)
- Shanghai Jiao Tong University.** *M.S. in Computer Science.* Mar 2015
Advisor: Prof. [Liqing Zhang](#)
- Shanghai Jiao Tong University.** *B.S. in Computer Science (ACM honored class).* May 2012
Advisor: Prof. [Yong Yu](#) and Prof. [Liqing Zhang](#)

EMPLOYMENT HISTORY

- Assistant Professor** [Michigan Technological University](#), Houghton, MI Aug 2025 – Present
Focus: Generative AI for Code and Software Engineering, Dept. of Computer Science
- Postdoctoral Fellow** [University of British Columbia](#), B.C., Canada Dec 2023 – Aug 2025
Focus: Generative AI for Code and Software Engineering with Prof. [Fatemeh Fard](#), Dept. of Computer Science
- Software Engineer** [Snap Inc.](#) (Snapchat), Seattle, WA Feb 2017 – Jul 2023
Teams: Friend Recommendation, Core Growth ML, User & Friend Infrastructure
- Software Engineer** [Microsoft Corporation](#), Redmond, WA Oct 2015 – Feb 2017
Team: OneNote Canvas & Ink
- Software Engineer** [ArcSite](#), Shanghai, China Mar 2015 – Oct 2015
Part of the startup team of five software engineers
- Research Intern** [Google Research](#), Shanghai, China Jun 2014 – Oct 2014
Mentor: Dr. [Hui Fang](#)
Project: Deep Learning for Image Enhancement
- Research Intern** [Microsoft Research Asia](#), Beijing, China Jul 2013 – May 2014
Team: Multimedia Search & Mining Group
Mentor: Dr. [Changhu Wang](#)
Project: Smart Ink
- Research Assistant** [Shanghai Jiao Tong University](#), Shanghai, China Jan 2012 – Jul 2013
Undergraduate & M.S. Research Study
Advisor: Prof. [Liqing Zhang](#), Dept. of Computer Science & Engineering
- Research Intern** [Microsoft Research Asia](#), Beijing, China Jul 2011 – Dec 2012
Team: Web Search & Data Mining Group
Mentor: Prof. [Ji-Rong Wen](#)
Project: Web Search Ranking for Numeric Queries

RESEARCH INTEREST

My research focuses on intent-aware software engineering, developing AI-assisted techniques to effectively align AI behavior with developer intent, while helping humans understand, guide, and verify AI-generated software to enable high-quality and safe software development.

Keywords: AI for Software Engineering (AI4SE), Software Engineering for AI (SE4AI), Agentic AI, Large Language Model (LLM) for Code.

PRE-PRINTS

- P1. Jie JW Wu.** [AssumptionMiner: Extracting, Tracing, and Revising Implicit Assumptions in LLM Code Generation.](#) *arXiv preprint arXiv:2607.22898*, 2026. (Under Review)

- P2.** Jie JW Wu, Manav Chaudhary, Davit Abrahamyan, Arhaan Khaku, Anjiang Wei, Fatemeh H. Fard. [Can Code Language Models Learn Clarification-Seeking Behaviors?](#) *arXiv preprint arXiv:2504.16331*, 2025. (Under Review)
- P3.** Shawn Zhao, Fatemeh H. Fard, Jie JW Wu. [MANTRA: a Framework for Multi-stage Adaptive Noise TReAtment During Training.](#) *arXiv preprint arXiv:2512.04319*, 2025. (Under Review).

REFEREED PUBLICATIONS

Journals

- J1.** Jie JW Wu*, Ayanda Patrick Herlihy*, Ahmad Saleem Mirza, Ali Afoud, Fatemeh Fard. [Growth-Hacker: Automated Off-Policy Evaluation Optimization Using Code-Modifying LLM Agents.](#) *ACM Transactions on Software Engineering and Methodology (ACM TOSEM)*, 2026. (*=Equal Contribution)
- J2.** Meghdad Dehghan, Jie JW Wu, Fatemeh H. Fard, Ali Ouni. [MergeRepair: An Exploratory Study on Merging Task-Specific Adapters in Code LLMs for Automated Program Repair.](#) *Empirical Software Engineering (EMSE)*, 2026 (Accepted; Registered Report accepted at ICSME 2024).
- J3.** Sathvik Joel*, Jie JW Wu*, Fatemeh H. Fard. [A Survey on LLM-based Code Generation for Low-Resource and Domain-Specific Programming Languages.](#) *ACM Transactions on Software Engineering and Methodology (ACM TOSEM)*, 2025. (*=Equal Contribution) (Presented as Journal-First paper at FSE 2026)
- J4.** Jie JW Wu, Fatemeh H. Fard. [Benchmarking the Communication Competence of Code Generation for LLMs and LLM Agent.](#) *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2025. (Presented as Journal-First paper at FSE 2026)
- J5.** Jie JW Wu, Thomas A. Mazzuchi, Shahram Sarkani. [A Multi-Objective Evolutionary Approach Towards Automated Online Controlled Experiments.](#) *Journal of Systems & Software*, 2023.
- J6.** Jie JW Wu, Thomas A. Mazzuchi, Shahram Sarkani. [Comparison of Multi-Criteria Decision-Making Methods for Online Controlled Experiments in a Launch Decision-Making Framework.](#) *Information and Software Technology*, 2023.

Conferences

- C1.** Jie JW Wu, Feiyu E, Bo Chen. [Metamorphic Testing for Clinical ML Models: A Framework Proposal and Pilot Study.](#) *ACM International Conference on AI-powered Software (AIware), Arxiv Track*, 2026.
- C2.** Mohamad Khajezade, Jie JW Wu, Fatemeh Hendijani Fard, Gema Rodríguez-Pérez, Mohamed Sami Shehata. [Investigating the Efficacy of Large Language Models for Code Clone Detection.](#) *International Conference on Program Comprehension (ICPC), Early Research Achievements (ERA)*, 2024.
- C3.** Jie JW Wu. [AutoOffAB: Toward Automated Offline A/B Testing for Data-Driven Requirement Engineering.](#) *Foundations of Software Engineering (FSE), Ideas, Visions and Reflections Track*, 2024.
- C4.** Jie JW Wu. [An Exploratory Study of V-Model in Building ML-Enabled Software: A Systems Engineering Perspective.](#) *3rd International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 2024 (**Distinguished Paper Award Candidate**).
- C5.** Jie Wu, Changhu Wang, Liqing Zhang, Yong Rui. [Offline Sketch Parsing via Shapeness Estimation.](#) *IJCAI 2015*. (full paper, acceptance rate: 28.6%).
- C6.** Jie Wu, Changhu Wang, Liqing Zhang, Yong Rui. [Sketch Recognition with Natural Correction and Editing.](#) *AAAI 2014*. (Demonstrated at **Microsoft TechFest 2015**) (full paper, acceptance rate: 28.0%).
- C7.** Jie Wu and Liqing Zhang. [Gestalt Saliency: Salient Region Detection Based on Gestalt Principles.](#) *IEEE International Conference on Image Processing (ICIP) 2013*. (B.S. Thesis, **Top 1% Excellent Undergraduate Thesis Award at SJTU**).

Workshop Papers (peer-reviewed)

- W1.** Rishab Sharma, Iman Saberi, Elham Alipour, Jie JW Wu, Fatemeh H. Fard. [FISCAL: Financial Synthetic Claim–document Augmented Learning for Efficient Fact-Checking.](#) *NeurIPS 2025 Workshop on GenAI in Finance*, 2025.

W2. Jie JW Wu. [Large Language Models Should Ask Clarifying Questions to Increase Confidence in Generated Code](#). *The 7th Annual Symposium on Machine Programming (MAPS' 23 Workshop)*, 2023.

PATENTS AND GRANTS

Patents

- Jonathan Brody, Donald Giovannini, Edward Koai, **Jie Wu**, Lin Zhong. “[Additive Friend Suggestion for Account Security](#)”. United States Patent No. US 11,171,972 B1 (Granted).
- Carine Ramses Iskander, **Jie Wu**, Ian William Mikutel, Sarah Elizabeth Sykes, David Glen Garber. “[Intuitive Selection of a Digital Stroke Grouping](#)”. United States Patent No. US 9,940,513 B2 (Granted).

TEACHING

Instructor, CS 5090 “Software Engineering with Generative AI”, grad course, Michigan Tech, Fall 2026

Instructor, CS 3712 “Software Quality Assurance”, undergrad/grad course for a class of ~60 students, Michigan Tech, Spring 2026

Instructor, CS 4321 “Introduction to Algorithms”, undergrad course for a class of 80 students, Michigan Tech, Fall 2025

Instructor, COSC 320 “[Analysis of Algorithm](#)”, undergrad course for a class of 80 students, University of British Columbia, Fall 2024

Instructional Skills Workshop (ISW) Certification, completion of a 24-hour Instructional Skills Workshop (ISW), which is recognized globally in higher education as a comprehensive program for enhancing teaching effectiveness, University of British Columbia, Fall 2024

Head Teacher, ACM Honored Class of 2016, Shanghai Jiao Tong University, Fall 2012 – Spring 2013

Teaching Assistant, “Practice on Programming and Coding Ability (PPCA)”, undergrad course for honored CS students, Shanghai Jiao Tong University, Fall 2011

ACADEMIC SERVICES

Software Engineering (SE) Curricula Committee Member, Computer Science Department, Michigan Tech

Publicity and Social-Media Co-chair, [CAIN 2027](#)

Program Committee Member, [ICSE 2027](#), Research Track

Program Committee Member, [CAIN 2027](#)

Reviewer, [ACM Transactions on Software Engineering and Methodology](#), 2025, 2026

Reviewer, [Communications of the ACM](#), 2026

Reviewer, [IEEE Transactions on Software Engineering](#), 2025

Reviewer, [IEEE Transactions on Dependable and Secure Computing](#), 2025

Reviewer, [Information and Software Technology](#), 2025

Program Committee Member, [AIware 2025](#), Benchmark and Dataset Track

Facilitator and Mentor, [Digital Research Infrastructure \(DRI\) Sensitization Workshop](#), UBC, Okanagan Campus

Program Committee Member, [RAIE 2025](#)

Program Committee Member, [CAIN 2025](#)

Program Committee Member, [SANER 2025](#), Tool and Demo Track

Reviewer, [Journal of Systems and Software](#), 2024, 2025

Program Committee Member, [ASE 2024](#), Artifact Evaluation Track

Program Committee Member, [FSE 2024](#), Posters Track

ADVISING AND MENTORING

Andrew Sadler (Summer 2026), Undergrad Student, Michigan Tech. *Role: Advisor*

Gabe Dautovi (Summer 2026), Undergrad Student, Michigan Tech. *Role: Advisor*

Joshana Shakya (Fall 2026, incoming), PhD Student, Michigan Tech. *Role: Advisor*
Indrajeet Roy (Fall 2026, incoming), PhD Student, Michigan Tech. *Role: Advisor*
Qiang Han (2025–Present), PhD Student, Michigan Tech (CS, Dr. Bo Chen’s group). *Role: Research Co-Mentor on a collaborative project*
Ayanda Patrick Herlihy (2025–Present), Undergrad Student at Birmingham City University, UK. *Role: Mentor*
Shawn Zhao (2025–Present), UBC PhD Student. *Role: Mentor*
Meghdad Dehghan (2024–Present), UBC Master (M.Sc.) Student. *Role: Mentor*
Ali Afoud (2025), UBC Undergrad Student. Working on COSC 448 Directed Studies in Computer Science. *Role: Co-supervisor*
Dabira Omotoso (2024–2025), UBC Undergrad Student. Working on Undergrad Honors Thesis. *Role: Co-supervisor*
Ahmad Saleem Mirza (2024–2025), UBC Undergrad Student. *Role: Mentor*
Manav Chaudhary (2024), Master Student at IIIT Hyderabad. Intern from Mitacs Program, UBC. *Role: Mentor*
Abhinav Thota (2023), UBC Undergrad. Published Undergrad Honors Thesis. *Role: Mentor*
Md Jumar Alam (2023), UBC Master (M.Sc.) Student. Published MSc Thesis. *Role: Mentor*

SELECTED TALKS

Guest presentation in CS 1000 (Michigan Tech) on the introduction of AI for coding and software engineering, 2025.
Talk at the Computer Science Colloquium at Michigan Tech, “Clarification-Aware Code Generation: Benchmarks, Techniques, and Open Problems”, 2025.
Talk at PNW PLSE 2025 at Uber Engineering Office in Seattle, WA, “Clarification-Aware Code Generation: Benchmarks and Techniques”, 2025.
Talk at Washington State University and Michigan Tech, “Building Trustworthy and Safe AIware With Systems Thinking”, 2025.
Presentation for Training Session at DRI sensitization workshop, University of British Columbia Okanagan, BC, Canada. Optimizing Large Language Model Workflows on ARC: A Practical Guide to Data Management, Inference, and Fine-tuning, 2024.
Talk at MAPS’ 23 Workshop, co-located with FSE in San Francisco, CA. Large Language Models Should Ask Clarifying Questions to Increase Confidence in Generated Code, 2023.
Talk at Learning Session, Friending team at Snap Inc., Seattle, WA. Introduction to Looker Dashboard and Data Analysis in Core Growth Team, 2022.
Talk at Learning Session, Friending team at Snap Inc., Seattle, WA. Introduction to Service Logging and Data Analysis at Friending Team, Core Growth Team, 2021.

AWARDS AND HONORS

- **Travel Grant Award**, FSE 2024 (USD \$1k) Jul 2024
- **Distinguished Paper Award Candidate**, CAIN 2024 Apr 2024
- **Travel Grant Award**, ICSE 2024 (USD \$1.2k) Apr 2024
- **Travel Grant Award**, FSE 2023 (USD \$1.7k) Apr 2024
- **“Engineer of the Month”** (two-time winner), Friending Team at Snap Inc. for contributions in Candidate Generation, Data, and Privacy of the Friending system Jan 2022
- **3rd place**, Infrastructure Cost Camp, Snap Inc. (USD \$800k/year cost saving) Jun 2020
- **Letter from CEO**, Snap Inc. in recognition of the new friend suggestion model Jun 2019
- **National Scholarship**, (Highest-level) Oct 2014
- **Excellent Master Academic Scholarship** (1st class), SJTU Nov 2013
- **Intel Fellowship** Sep 2013

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| • Top 1% Excellent Undergraduate Thesis , SJTU (38 out of 4000 theses) | Jun 2012 |
| • Microsoft Fellowship for Undergraduates (30 recipients in China) | Aug 2011 |
| • First Class Prize , National Olympiad in Informatics in Provinces (NOIP) | Oct 2007 |

Last updated: July 2026.